

TraLO: the promising result, in context

Three-page meeting brief | 1 October 2026 | Development-set evidence

Our best supported finding is modest: on knee X-rays with MobileNetV3-Large, TraLO places about one more true grade-3 case among 76 available slots than the same model with the same post-training Clipper. This is a 72-seed result. It is not evidence that TraLO works on every dataset or backbone.

1. What are we trying to improve?

Suppose a clinic can send only **76 of 826** knee X-rays to the grade-3 treatment pathway. First a network assigns a score to each image. Then a **Clipper** enforces the limit by reserving the 76 highest grade-3 scores. The clinic cares about which patients occupy those slots, not just whether the number is exactly 76.

The headline measure is **constrained-class F1 (cc-F1)** for grade 3 after the Clipper. It balances missed true cases and incorrect referrals. We also track accuracy, macro-F1 across all five grades, and weighted-F1, which gives common grades more influence. A gain in cc-F1 accompanied by damage to all other grades would need careful judgment.

2. Where is TraLO, and where is Kassif's method?

Step	Kassif's PAO	Our TraLO study
Supervised base	Train an image network on labeled knee images.	We adopted and tested his stronger training recipe: image augmentation, class-balanced sampling and early stopping.
Constraint update	PAO repeatedly changes the cost-sensitive training loss, up-weighting labeled training errors that overfill a class, then retrains.	TraLO keeps the supervised run fixed and takes a small gradient step on each saved model snapshot using predicted counts from unlabeled deployment images.
Final decision	Apply the capacity-aware Clipper.	Apply the same Clipper to TraLO and all controls; average snapshot probabilities before the cut.

So we did **not** simply rename PAO. The architecture can be the same; the scientific difference is the constraint update. The stronger base training recipe improved our old pipeline substantially, even without TraLO. The experiment below isolates TraLO's additional contribution over that stronger baseline.

3. The positive comparison: knee X-rays, MobileNetV3-Large

The 72 seeds use the same images, model training, saved snapshots and Clipper. Only the TraLO direction changes. Percentages are means across seeds; +/- is between-seed standard deviation (SD). 'pp' means percentage points.

Measure	Plain snapshots + Clipper	TraLO snapshots + Clipper	Paired TraLO gain, 95% CI
Grade-3 cc-F1	73.28 +/- 1.83%	74.45 +/- 1.69%	+1.18 pp [+0.84, +1.51]
Overall accuracy	See paired contrast	See paired contrast	+0.87 pp [+0.63, +1.11]
Macro-F1	See paired contrast	See paired contrast	+0.93 pp [+0.67, +1.19]
Weighted-F1	See paired contrast	See paired contrast	+0.46 pp [+0.23, +0.69]

The full score file reports paired differences for the three secondary metrics, not the two absolute arm averages, so the table gives the audited differences rather than inventing missing averages. The same-size **random sham step** adds essentially nothing to cc-F1; TraLO also beats that sham by +1.18 pp [+0.85, +1.50]. The prespecified primary contrasts pass their adjusted significance test.

What does a 1.18-point improvement mean for patients?

Mean grade-3 decision counts	True grade 3	Not grade 3
Plain / selected	66.7 correct	9.3 incorrect
Plain / not selected	39.3 missed	710.7 correctly excluded
TraLO / selected	67.8 correct	8.2 incorrect
TraLO / not selected	38.2 missed	711.8 correctly excluded

This is a **derived mean 2-by-2 matrix**, not one seed's raw confusion matrix: each run selects 76 cases, the development pool has 106 true grade-3 cases, and the mean cc-F1 determines mean true positives. The effect is roughly **one extra correct patient in the same 76 slots**.

A separate local-constraint result is not yet a win

Satellite MobileNetV3, cap 167	Plain + Clipper	Local TraLO + Clipper
Class-1 cc-F1	49.17%	49.56%
Accuracy / macro-F1 / weighted-F1	54.12/47.73/55.85	52.52/46.31/54.41

The cc-F1 change is only +0.39 pp, with a 95% interval from -0.19 to +0.98; accuracy and the broader F1 scores fall. We therefore do not call this a positive transfer result.

4. Kassif's PAO beside TraLO: our matched re-run

The table below is **our own 24-seed ResNet18 reproduction of Kassif's training pipeline** on the same knee development pool and grade-3 cap, with his PAO, plain predict-then-optimize (PTO), and our TraLO side-step. It is **not a numerical table copied from Kassif's paper**.

Method in our re-run	cc-F1	Accuracy	Macro-F1	Weighted-F1
Plain PTO + Clipper	69.78%	62.45%	62.83%	61.25%
Kassif PAO + Clipper	69.69%	61.38%	62.63%	60.58%
TraLO step + Clipper	70.51%	62.61%	62.65%	60.89%

Bold marks the highest point estimate within this table, not a proven overall winner. Across seeds, PAO minus plain PTO is -0.09 cc-F1 points [-2.85, +2.66]; TraLO minus its same-size random sham is +0.73 [+0.08, +1.38], but misses the prespecified multiple-comparison threshold (adjusted $p = 0.088$). The paired difference SDs are 6.52 and 1.54 points, respectively. This smaller-backbone result is suggestive, not decisive.

What did Kassif's article itself report?

Kassif and Singer describe PAO as outperforming a predict-then-optimize pipeline on knee images and DermaMNIST, especially under tight capacity. The accessible publisher preview does **not** expose its numerical results tables, and its setup is not identical to our matched re-run. We therefore cannot honestly place claimed article percentages in the table above. The source is DOI 10.1016/j.engappai.2026.115989; the author's code is github.com/YuvalKassif/ConstrainedClassification.

5. Is this enough for a paper?

A lead, not a finished claim. TraLO is mathematically distinct from PAO and has a reproducible 72-seed knee MobileNetV3 gain over a strong Clipper and a random-step control. But the gain is small, the development set has been viewed repeatedly, and local satellite results are inconclusive. A general or clinical superiority claim would be premature.

The first next experiment is a **matched full-training comparison** of TraLO, PAO/ALM, plain Clipper and a zero-update null on one frozen modern backbone and the same compute budget. That experiment is prepared but has no accepted result yet. Second, study the **ranking** of cases entering limited slots: lowering a predicted class count is less valuable when the Clipper already enforces the count. Third, freeze the method and evaluate once on untouched cases; the sealed test sets remain unopened.

For the professor: our practical training pipeline improved mostly because of better image training and snapshot averaging borrowed and tested from Kassif's setup. TraLO adds a separate, small knee benefit on MobileNetV3. If the matched comparison and untouched evaluation fail to preserve that benefit, we should keep the stronger base classifier and stop pursuing this count-gradient version of TraLO.

Audit sources: experiments/claude_stepens_additional_result_20260928.md; experiments/claude_yuval_pipeline_result_20260927.md; experiments/fmow_boundary_mnv3_result_20261001.md; experiments/tralo_pdf_evidence_20261001.json. No reserved-country or sealed knee labels were used.